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Effects of Permanent Separation from Mother on Infant Monkeys[1]

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Rhesus monkey infants were reared from birth with their mothers in a common pen, then separated permanently from their mothers at 60, 90, or 120 days of age. Half of the subjects in each age group were housed individually following separation; the other half were housed in pairs. Although the subjects did not differ appreciably in levels of behaviors scored prior to separation, their behaviors both in the week following separation and at six months of age varied according to age of separation and subsequent housing condition manipulations. All monkeys showed agitation immediately following separation. Monkeys separated at 90 days of age showed quantitatively more severe immediate reaction to separation but by six months of age did not differ appreciably from monkeys separated at other ages. Monkeys housed singly following separation showed significantly higher levels of disturbance and self-clasping behavior and lower levels of locomotion during the week following separation than monkeys housed in pairs. These differences were exaggerated at six months of age.

Phenomena resulting from disruption of strong social attachment bonds have been extensively studied by ethologists, by comparative, developmental, and social psychologists, and by clinical psychologists and psychiatrists. Such research has most commonly taken the form of observing the behavioral consequences of separation of infant from mother or vice versa. Both naturally occurring separations and those imposed deliberately by experimenters have been studied across a variety of species, including avian, ungulate, canine, and human subjects.

Of recent interest have been the numerous research studies involving deliberate separation and subsequent reunion of macaque mothers and infants. Virtually all of the authors have reported that the infant's initial reaction to physical separation from the mother is one of agitation, involving increases over preseparation levels in activity, usually seen in locomotion and vocalization. In some species, this is most often followed by a period of inactivity, usually characterized by social withdrawal and exhibition of passive, self-directed behaviors. Reunion with the mother is typically marked by a temporary increase in mother–infant activity over preseparation levels. In some cases, effects of the separation persist indefinitely.

Kaufman and Rosenblum (1967) have interpreted the biphasic reaction seen in pigtail macaque infants as indicative of an adaptive conservation-withdrawal mechanism for energy preservation and avoidance of detection by potential predators. The same investigators (Kaufman & Rosenblum, 1969) have reported marked species differences in infant reaction to maternal separation, which they believe reflect species differences in macaque maternal practices. Hinde and his co-workers (e.g., Hinde & Spencer-Booth, 1971; Hinde, Spencer-Booth, & Bruce, 1966) have used the mother–infant separation paradigm to index the development of dynamic mother–infant interrela-

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tionships, as well as to highlight individual differences among mother–infant pairs. Researchers at the University of Wisconsin Primate Laboratory (e.g., Harlow, Suomi, & McKinney, 1970) have employed the paradigm for simulation of human anaclitic depression, first reported by Spitz (1946), Bowlby (1960), and others to result from separation of human mother and child. This approach has also been suggested by several others in the field of animal models of human syndromes (e.g., McKinney & Bunney, 1969; Senay, 1966).

In the course of the above investigations, numerous parameters regarding mother–infant separation have been examined. Age of infant at separation has not been found to influence qualitatively the nature of reaction to separation, although the range of ages investigated has been limited to only the middle months of the infant's first year of life. On the other hand, large inter- and intraspecies differences in maternal care of the infant prior to separation have yielded corresponding differences in the nature of the infant's responses to both separation and subsequent reunion. Duration of separation has been found generally to correlate positively with severity of infant reaction to separation. Also, the nature of the environment for both mother and infant during the period of separation has been shown to affect responses both during separation and following reunion. These findings are summarized in reviews by Hinde and Spencer-Booth (1971) and by McKinney, Suomi, and Harlow (1973).

The present experimentation was designed to investigate the effects of *permanent* separation from the mother among rhesus monkey infants of varying ages. Subjects were separated from their mothers at 60, 90, and 120 days of age and were subsequently housed either individually or in pairs.

Method

Subjects

The subjects were 12 rhesus monkeys (*Macaca mulatta*), 6 males and 6 females. Following birth, each subject was housed with its mother in a 19.5 × 7.5 × 7 foot group living pen containing other mothers and infants and was assigned to one of three groups: 60-day separation, 90-day separation, or 120-day separation. Each group consisted of 2 male and 2 female subjects. Each subject remained with its mother until separation.

Procedure

When each member of the 60-day group reached 60 days of age, it was separated physically from its mother. Separation was achieved by first catching the mother and the infant in a handler's net, then restraining the mother while wrestling the infant from her grasp. The entire procedure usually took less than 30 seconds to complete. Following separation, one male subject and one female subject were housed singly in 2.5 × 2 × 2.5 foot wire-mesh cages. The remaining male and female subjects were housed together in a 5 × 2 × 2.5 foot wire-mesh cage. All four subjects were housed in the laboratory in a different room on a different floor from where their mothers were housed.

An identical procedure was followed at 90 days of age for members of the 90-day group and at 120 days of age for members of the 120-day group. Members of all three groups remained in their respective cages until 180 days of age.

Data Collection

Each subject was observed for a five-minute period four days per week throughout its first six months of life. A single observer recorded the frequency and duration of behaviors falling into predefined categories on a bank of clocks and counters. Operational definitions of the categories are the following:[3] (*a*) *mother:* ventral–ventral contact between mother and infant or infant sitting within the embracing arms of the mother; (*b*) *mom huddle:* any other physical contact with infant's own mother or orientation toward own mother within one body length; (*c*) *peer huddle:* ventral–ventral or ventral–dorsal contact with another infant (the infant being scored on this behavior would have to have his ventral surface to the other infant); (*d*) *peer play:* brief contact play, rough-and-tumble, or pursuit–avoidance play; (*e*) *peer explore:* tactile or oral manipulation of another infant; (*f*) *active disturbance:* screeching or loud prolonged cooing while locomoting or sterotypy (repetition of a locomotor pattern for at least three cycles); (*g*) *passive disturbance:* screeching or loud cooing while stationary or rocking; (*h*) *infant reject:* negative behavior shown toward another infant such as threat, aggression, or fear grimace; (*i*) *other-mom contact:* any physical contact with another mother or orientation toward her within one body length;

[3] The self-clasp-huddle category was not scored prior to maternal separation. The mother, mom-huddle, and other-mom-contact categories were not scored following maternal separation. The peer-huddle, peer-play, and peer-explore categories were scored only for monkeys housed in pairs.

(*j*) *self-clasp huddle:* prolonged holding of some body part with hands or feet other than for support of that body part and/or bending top portion of body downward while tucking head into chest, giving the appearance of curling into a ball; (*k*) *locomotion:* movement by animal of at least one body length; and (*l*) *environment explore:* tactile or oral manipulation of the cage. Observations took place between the hours of 10:00 a.m. and 2:30 p.m. throughout the study. Order of testing among subjects was determined randomly.

Data Analysis

The duration data from this study were analyzed in three stages. First, behaviors of all subjects prior to separation were compared in order to determine if possible preseparation differences might be distributed unevenly among subsequent subject groups. This was accomplished by calculating for each subject a mean for each behavioral category recorded during the first 30–60 days of life. The means for every category were then subjected to a two-way analysis of variance, with separation group (60-day, 90-day, or 120-day) and housing condition (housed singly or together) as the uncorrelated variables. Statistical significance for these and subsequent tests was arbitrarily set at $\alpha = .05$.

Second, the subjects' initial reactions to separation were compared. For each category, scores recorded during the week following separation were subjected to a three-way repeated-measures analysis of variance, with separation group and housing condition as the uncorrelated variables and days following separation (Days 1–2 versus Days 3–7) as the repeated measure. For those categories which yielded significant main effects and/or interactions, a Duncan new multiple-range test (Duncan, 1955) was applied subsequently to the appropriate means.

Finally, long-term effects of the separations were examined for possible systematic variance within age of separation and/or subsequent housing variables. For every subject, scores recorded for nonsocial behaviors during Days 150–180 of life were summed and divided to yield a mean for each nonsocial category. Each set of means was then subjected to a two-way analysis of variance, with separation group and housing condition as the uncorrelated variables. For subjects housed in pairs, similarly obtained scores recorded on social behaviors were analyzed by one-way analyses of variance, with separation group as the uncorrelated variable. Duncan multiple-range tests were subsequently applied as above.

Results

The results of the present experiment indicated that although subjects' behaviors did not differ appreciably prior to maternal separation, significant differences related to age and housing variables emerged following separation. Generally, subjects separated at 90 days of age exhibited a more severe immediate reaction to separation than other subjects, although this difference was not evident on long-term measures. A second major finding was that animals housed singly showed far more severe reactions to separation, both immediate and long term, than animals housed in pairs following separation.

Analysis of variance of subject scores prior to separation failed to yield any significant main effects or interactions for any of the categories examined. In other words, there was no evidence in the second month of life that subjects differed systematically in levels of behaviors across the preassigned age or housing groups. In addition, in order to verify that there was nothing unusual about mother–infant interactions prior to separation, another calculation was made. For each chronological two-week period, a mean for each category of mother-directed behavior was computed for those subjects still with their mothers. These means are plotted in Figure 1 and are not noticeably different from similarly obtained data from previous studies of mother–infant interactions, for example, Harlow, Harlow, and Hansen (1963) and Hansen (1966).

In general, all subjects showed agitation immediately following separation. However, the analyses of postseparation behavioral levels yielded consistent age, housing condition, and days following separation differences across several categories. Mean levels of active disturbance behaviors are shown in Figure 2. Analysis of variance of active disturbance scores yielded a significant main effect of days following separation (Days 1–2 > Days 3–7). Significant Housing Condition × Days and Age × Housing Condition × Days interactions were also found, with singly housed subjects separated at 90 days of age showing higher levels during Days 1–2 of separation than any other subject groups during either time period. Figure 3 illustrates mean levels of passive disturbance behaviors. Analysis of variance of these scores yielded a significant effect of days (Days 1–2 > 3–7) and a significant three-way interaction, resulting primarily from elevated levels of the 90-day singly housed group during Days 1–2 of separation. Mean levels of self-

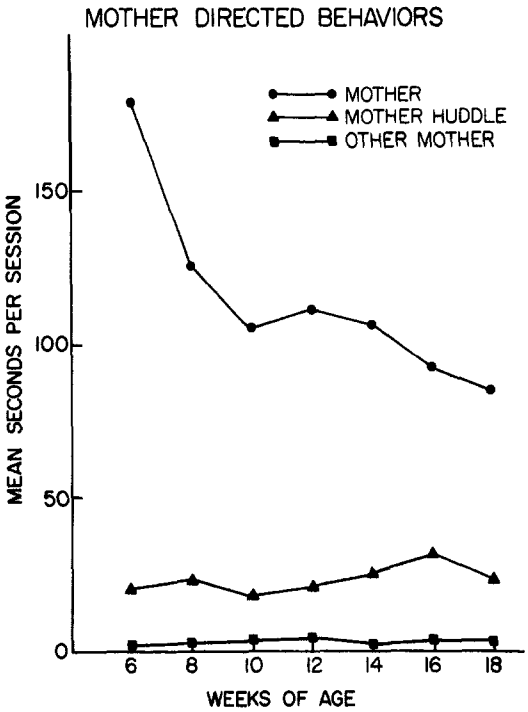


FIG. 1. Mean subject levels of mother-directed behaviors.

clasp huddle are plotted in Figure 4. The analysis of variance yielded significant main effects of all three variables (90-day > 60-day and 120-day groups singly housed > housed together, and Days 1–2 < Days 3–7), a significant Age × Housing condition interaction (90 days singly housed > all other means), and a significant Age × Day interaction (90-day group during Days 3–7 > all other means). Analysis of variance of the locomotion scores yielded a significant effect of housing (housed together > singly housed) and a significant Housing × Days interaction (singly housed during Days 3–7 < all other means). Analysis of variance of environmental exploration scores failed to yield any significant main effects or interactions. To summarize these analyses, all subjects reacted to separation with agitation during Days 1–2 with the most extreme form shown by 90-day singly housed subjects. Those subjects housed singly subsequently exhibited behavior patterns comparable to the "despair stage" of reaction to separation reported by other investigators (e.g., Kaufman & Rosenblum, 1967).

Long-term effects of the separations and subsequent form of housing are illustrated in Figures 5 and 6. Analysis of variance of locomotion scores yielded only a significant effect of housing condition (housed together > singly housed), while the analyses for the categories of environmental exploration, active disturbance, and passive disturbance failed to yield any significant main effects or interactions. On the other hand, analysis of variance of the self-clasp-huddle scores yielded a significant effect of age (120-day > 60-day and 90-day groups), housing (singly housed > housed together), and a significant interaction (120-day singly housed > 60-day singly housed and 120-day singly housed > all groups housed together). The analyses of social behavior scores for the subjects housed in pairs revealed unsystematic age differences across behaviors. Subjects separated at 60 days showed less peer huddle and more peer play than those separated at 90 or 120 days, while the 120-day monkeys showed higher levels of peer explore. These differences just barely reached acceptable levels of statistical significance. In general, long-term effects of separation across all behaviors were more influenced by the nature of subsequent housing than by the age of separation.

Discussion

The results of the present experimentation clearly indicated that differences in both age of separation and subsequent housing condition were associated with differences in infant monkey behaviors following maternal separation. These findings have a number of implications for theoretical and practical aspects of mother–infant separation which merit discussion.

The present data failed to disclose major qualitative differences in reaction to separation among monkeys separated at 60, 90, or 120 days of age. Quantitative differences in immediate reaction were found, however, with those monkeys separated at 90 days generally exhibiting a more severe reaction than all other subjects. While other research has failed to obtain gross age differences when examining subjects under a year of age (Abrams, 1969; Hinde & Spencer-Booth, 1971), the present data are of in-

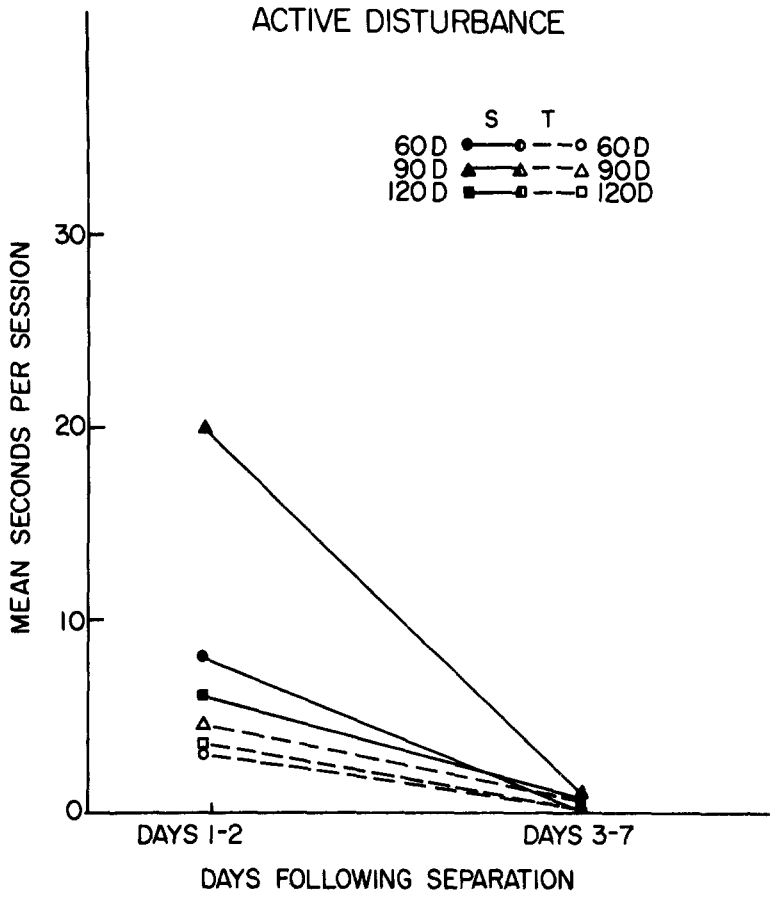


FIG. 2. Mean levels of active disturbance following separation.

terest in terms of the age range examined. Most previous studies included subjects at least four months of age at time of separation. The present study included animals that were separated prior to maturation of the fear response, which in rhesus monkeys normally occurs at 70–90 days of age (Harlow & Harlow, 1965; Sackett, 1966). Maturation of the corresponding response in human infants (Freedman, 1961; Spitz, 1950) has been posited by some to be associated with a qualitative change in form of object of infant attachment from general to specific (Shaffer & Emerson, 1964). Others have maintained that the two phenomena are not necessarily causally related (Morgan & Ricciuti, 1969). We believe that the present data support the latter position, in that qualitative differences in reaction to separation between the 60-day and the other subjects would be predicted if a causal relationship did exist, but no such differences were disclosed. On the other hand, quantitative differences between the 90-day and the other subjects immediately following separation were apparent, consistent with previous findings that immediately following maturation of the fear response, infant monkeys are more adversely affected by gross environmental changes than at times earlier or later in life.

The present data indicated that housing condition following separation influenced the monkeys' responses to separation. Although all subjects showed clear evidence of a "protest" reaction during the first two days of separation, singly housed subjects, particularly those separated at 90 days, tended to exhibit higher levels of disturbance activity. More importantly, singly housed subjects

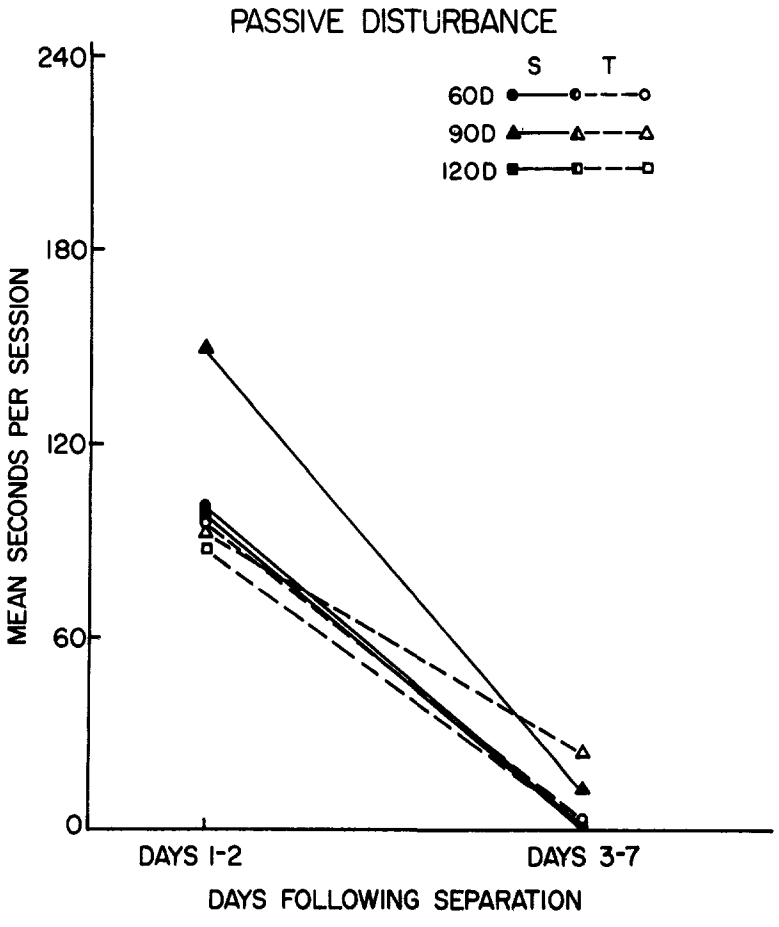


FIG. 3. Mean levels of passive disturbance following separation.

showed much higher levels of self-clasping behaviors and much lower levels of locomotor behavior during the rest of the week following separation than subjects housed in pairs. The results of the present study are thus consistent with previous separation studies: Virtually all separated monkeys show protest, but only some subsequently exhibit clear-cut "despair" reactions to maternal separation. This differential reaction has previously been explained in terms of preseparation variables such as permissiveness of mother (Kaufman & Rosenblum, 1967), availability of potential substitute mothers (Hinde & Davies, 1972; Kaufman & Rosenblum, 1969; Schlottmann & Seay, 1972), and strength and nature of the mother–infant bond (Hinde & Spencer-Booth, 1971). The present data clearly demonstrated that postseparation variables such as housing condition may also be crucial. It is not clear whether the present paired subjects failed to show high levels of self-clasping and low levels of locomotion because their partners provided comfort and security or because their partners provided stimulation to do something besides self-clasp, for example, play or cling to partner. At any rate, with regard to exhibition of despair behaviors, housing condition was a variable of importance among the present subjects.

Housing condition was also a significant variable when long-term effects of the separations were assessed. Singly housed subjects again showed higher levels of self-clasp and lower levels of locomotion than subjects housed in pairs. In fact, inspection of the data indicates that singly housed subjects'

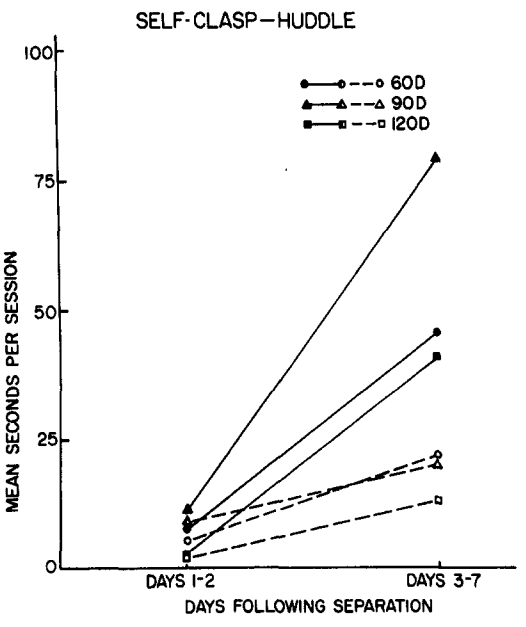


FIG. 4. Mean levels of self-clasp huddle following separation.

levels of self-clasping were higher and their levels of locomotion lower than they had been at the end of the week immediately following separation. On the other hand, by six months of age subjects housed together showed increases in locomotion over levels recorded the week following separation, and the levels of their social scores did not differ much from those of well-socialized monkeys of similar age who in previous studies had not been subjected to maternal separation

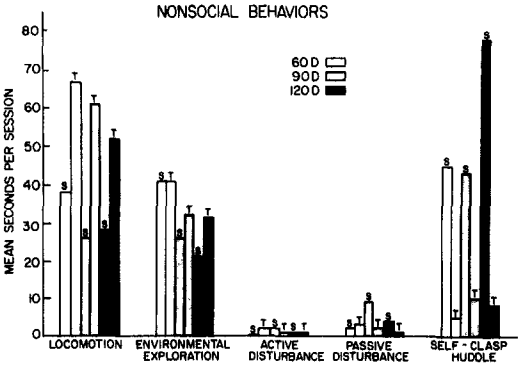


FIG. 5. Mean levels of nonsocial behaviors, 150–180 days of age.

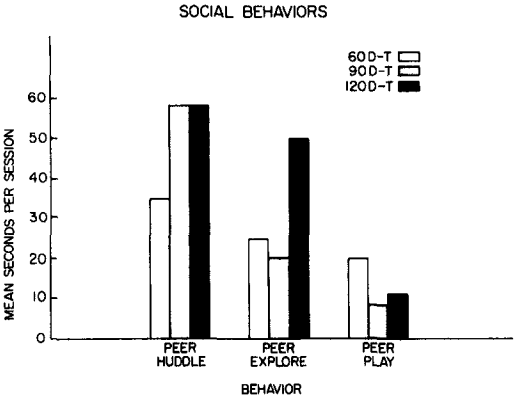


FIG. 6. Mean levels of social behaviors, 150–180 days of age.

(e.g., Chamove, 1966; Hansen, 1966; Rosenblum, 1961; Suomi, 1973). Indeed, socially they have performed adequately in experimentation subsequent to the present study. One is tempted to conclude from the present study that the reaction to separation is intensified by subsequent individual housing but mollified by subsequent social housing, a conclusion reached long ago by Spitz (1946) from study of human infants.

However, an alternative interpretation may account in part for the obtained data. Numerous investigators have found that monkeys reared in single cages *from birth* exhibit behavioral levels at six months of age not unlike the singly housed subjects in the present study (Cross & Harlow, 1965; Harlow & Harlow, 1969; Pratt, 1969; Sackett, 1965; Suomi, Harlow, & Kimball, 1971). Hence, it is conceivable that the present subjects' behaviors at six months were not so much a result of the maternal separation per se but instead a result of a particular type of housing condition. At any rate, it is clear that the behavior of the present subjects following separation was influenced by the nature of their subsequent home environment.

The present data may thus be added to a growing body of literature which suggests, as McKinney et al. (1973) have pointed out, that separation becomes a meaningless predictive concept unless certain parameters describing the particular subjects and situation involved in separation are delineated. These data raise points to be considered by animal and human social researchers alike. At the

very least they should caution an investigator from making global inferences regarding human reaction to social separation from a single study of monkey reaction to social separation. At the very most they should encourage human researchers to consider carefully environmental contingencies when pondering the consequences of disruption of the mother–infant bond.

REFERENCES

ABRAMS, P. S. Age and effect of separation on mother and infant rhesus monkeys *(Macaca mulatta)*. Paper presented at the meeting of the American Association of Physical Anthropologists, Mexico City, 1969.

BOWLBY, J. Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child,* 1960, **15,** 9–52.

CHAMOVE, A. S. The effects of varying infant peer experience on social behavior in the rhesus monkey. Unpublished master's thesis, University of Wisconsin, 1966.

CROSS, H. A., & HARLOW, H. F. Prolonged and progressive effects of partial isolation on the behavior of macaque monkeys. *Journal of Experimental Research in Personality,* 1965, **1,** 39–49.

DUNCAN, D. B. Multiple range and multiple *F* tests. *Biometrics,* 1955, **11,** 1–42.

FREEDMAN, D. G. The infant's fear of strangers and the flight response. *Journal of Child Psychology and Psychiatry,* 1961, **2,** 242–248.

GRIFFIN, G. A., & HARLOW, H. F. Effects of three months of total deprivation on social adjustment and learning in the rhesus monkey. *Child Development,* 1966, **37,** 533–547.

HANSEN, E. W. The development of maternal and infant behavior in the rhesus monkey. *Behaviour,* 1966, **27,** 107–149.

HARLOW, H. F., & HARLOW, M. K. The affectional systems. In A. M. Schrier, H. F. Harlow, & F. Stollnitz (Eds.), *Behavior of nonhuman primates.* Vol. 2. New York: Academic Press, 1965.

HARLOW, H. F., & HARLOW, M. K. Effects of various mother-infant relationships on rheus monkey behaviors. In B. M. Foss (Ed.), *Determinants of infant behaviour.* Vol. 4. London: Methuen, 1969.

HARLOW, H. F., HARLOW, M. K., & HANSEN, E. W. The maternal affectional system of rhesus monkeys. In H. L. Rheingold (Ed.), *Maternal behavior in mammals.* New York: Wiley, 1963.

HARLOW, H. F., SUOMI, S. J., & MCKINNEY, W. T. Experimental production of depressive behaviors in young monkeys. *Mainly Monkeys,* 1970, **1,** 6–12.

HINDE, R. A., & DAVIES, L. Removing infant rhesus from mother for 13 days compared with removing mother from infant. *Journal of Child Psychology and Psychiatry,* 1972, **13,** 227–237.

HINDE, R. A., & SPENCER-BOOTH, Y. Effects of brief separation from mother on rhesus monkeys. *Science,* 1971, **173,** 111–118.

HINDE, R. A., SPENCER-BOOTH, Y., & BRUCE, M. Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature,* 1966, **210,** 1021-1033.

KAUFMAN, I. C., & ROSENBLUM, L. A. The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine,* 1967, **29,** 648–675.

KAUFMAN, I. C., & ROSENBLUM, L. A. The waning of the mother-infant bond in two species of macaque. In B. M. Foss (Ed.), *Determinants of infant behaviour.* Vol. 4. London: Methuen, 1969.

MCKINNEY, W. T., & BUNNEY, W. E. Animal model of depression. *Archives of General Psychiatry,* 1969, **21,** 240–248.

MCKINNEY, W. T., SUOMI, S. J., & HARLOW, H. F. New models of separation and depression in rhesus monkeys. In E. Senay & J. P. Scott (Eds.), *Separation and depression: Clinical and research aspects,* 1973, in preparation.

MORGAN, G. A., & RICCIUTI, H. N. Infants' responses to strangers during the first year. In B. M. Foss (Ed.), *Determinants of infant behaviour.* Vol. 4. London: Methuen, 1969.

PRATT, C. L. The developmental consequences of variations in early social stimulation. Unpublished doctoral dissertation, University of Wisconsin, 1969.

ROSENBLUM, L. A. The development of social behavior in the rhesus monkey. Unpublished doctoral dissertation, University of Wisconsin, 1961.

SACKETT, G. P. Effects of rearing conditions upon the behavior of rhesus monkeys. *Child Development,* 1965, **36,** 855–868.

SACKETT, G. P. Monkeys reared in isolation with pictures as visual input: Evidence for an innate releasing mechanism. *Science,* 1966, **154,** 1468–1472.

SCHLOTTMANN, R. S., & SEAY, B. Mother-infant separation in the Java monkey *(Macaca irus)*. *Journal of Comparative and Physiological Psychology,* 1972, **79,** 334–340.

SENAY, E. C. Toward an animal model of depression: A study of separation behavior in dogs. *Journal of Psychiatric Research,* 1966, **4,** 65–71.

SHAFFER, H. R., & EMERSON, P. E. Development of social attachments in infancy. *Monographs of the Society for Research in Child Development,* 1964, **29,** 1–77.

SPENCER-BOOTH, Y., & HINDE, R. A. Effects of 6 days separation from mother on 18- to 32-week-old rhesus monkeys. *Animal Behaviour,* 1971, **19,** 174–191.

SPITZ, R. A. Anaclitic depression. *The Psychoanalytic Study of the Child,* 1946, **2,** 313–347.

SPITZ, R. A. Anxiety in infancy: A study of its manifestations in the first year of life. *International Journal of Psychoanalysis,* 1950, **31,** 138–143.

SUOMI, S. J. Repetitive peer separation of young monkeys: Effects of vertical chamber confinement during separations. *Journal of Abnormal Psychology*, 1973, **81**, 1–10.

SUOMI, S. J., HARLOW, H. F., & KIMBALL, S. D. Behavioral effects of prolonged partial social isolation in the rhesus monkey. *Psychological Reports*, 1971, **29**, 1171–1177.

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